

Smart Traffic Lights using Local AI

Student Handout – AI-Assisted Research & L^AT_EX Workshop

Abstract—Smart traffic-light systems are being studied to reduce waiting times and adapt rapidly to dynamic road conditions. This paper provides a literature review comparing traditional fixed-timer traffic signals, cloud-based AI systems, and local edge-based smart cameras. We outline a standard four-step system architecture, evaluate system performance metrics, and highlight key research gaps regarding emergency vehicle prioritization.

Index Terms—Artificial Intelligence, Edge Computing, IEEE Style, Smart Traffic Lights, Traffic Signal Control.

I. INTRODUCTION

Traditional traffic lights usually follow fixed timers. While simple and inexpensive, they cannot easily react when a single road experiences sudden congestion. Modern adaptive systems leverage cameras and artificial intelligence (AI) to estimate vehicle density and dynamically adjust signal timing. Instead of relying strictly on clock cycles, decisions are informed by real-time roadway metrics. Recent reviews confirm a growing interest in AI-based traffic signal control and its potential benefits under changing conditions [1].

II. LITERATURE REVIEW

A. Cloud vs. Local Processing

Cloud-based AI systems transmit camera feed data to a central server for processing. Although this offers high computing capacity, it introduces dependency on stable internet connectivity and increases communication latency.

Conversely, local smart-camera (edge computing) systems execute processing near the roadway. This minimizes bandwidth consumption and supports faster decision-making. Small local computers analyze localized camera feeds to issue immediate instructions to the signal controller. Machine-vision research highlights the efficiency of camera-based adaptive signal modes [2].

B. Critical Evaluation of Microcontroller Capabilities

Literature indicates extreme claims regarding low-cost microcontrollers. For example, Smith et al. [3] presented a claim where a basic Arduino Uno processed 10,000 live HD video streams simultaneously with zero delay. When evaluating such literature, it is essential to distinguish between actual hardware computational limits (such as 8-bit, 16 MHz hardware constraint boundaries) and theoretical or simulated claims.

III. SYSTEM ARCHITECTURE

The proposed system architecture is structured into four sequential steps:

- 1) **Street Cameras:** Capture live traffic conditions on-site.
- 2) **Smart Pole Computers:** Perform localized video processing and issue rapid traffic management decisions.

- 3) **City Cloud Server:** Receives selected summarized data for long-term storage and system-wide analytics.
- 4) **Traffic Lights:** Receive actuation decisions to adjust signal indicators.

The data and decision flow is summarized as follows:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

IV. PERFORMANCE COMPARISON

Table I summarizes the three primary traffic light system paradigms based on response time, bandwidth demand, and installation costs.

TABLE I
COMPARISON OF TRAFFIC SYSTEM ARCHITECTURES

System Type	Decision Time	Internet Data	Setup Cost
Basic Timed Lights	30.0 s	Zero	\$200
Cloud AI Systems	2.0 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

V. RESEARCH GAP

Existing traffic signal control solutions function effectively under regular, predictable daily commuting patterns. However, a significant research gap persists in emergency scenarios: many existing systems lack integrated protocols to automatically clear paths for approaching emergency vehicles, such as speeding ambulances. There is a distinct need for smart traffic architectures capable of detecting priority vehicles and dynamically coordinating surrounding intersections to route them safely.

REFERENCES

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